ENVIRONMENTAL PRODUCT DECLARATION



Teodorico



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1. The company and the product

San Marco Group SpA is working on the quantification of the environmental impacts of its products in order to improve their technical and environmental performance.

1.1 San Marco Group S.p.A.

San Marco Group is an international group, operating in the production and marketing of paints and varnishes for professional construction in Italy. The headquarters are located in Marcon, Venice, where part of the production takes place, together with the innovation centre and two training centres. The Group consists of 7 production sites and 3 sales companies located in Europe, where more than 300 people work directly. International growth has allowed the Group to be present in most parts of the globe with a network of distributors touching more than 100 countries on five continents, with a widespread network of sales outlets.



The Group manages 6 different brands. The key factors for San Marco Group are the search for opportunities in foreign markets, the acquisition of other entities and the strengthening of the services offered.

The Group's various brands, acquired or developed internally, have expanded business opportunities into new markets and new customer segments. All of the Group's companies have a precise role to play, so as to enhance their specificities and define their positions, reducing overlaps and exploiting synergies. This type of organisation has progressively led to the upgrading of machinery and plants, increasing the quantity and quality of products, all through a commitment to constant improvement and growth.

San Marco Group has voluntarily chosen to establish, implement, maintain and improve its integrated management system for quality, the environment and the health and safety of workers.

Indeed, San Marco Group obtained the ISO 9001:2015 Quality Certification through the Certification Body DNV-GL, for all the Italian production unit of San Marco Group S.p.A. (Marcon, Latisana and Montemarciano), for the company of Forlì, Novacolor S.r.l., for the Bosnian company San Marco DT BH Doo, and for the Norwegian company Tjaeralin AS. It obtained the ISO 14001:2015 Environmental Certification again with the Certification Body DNV-GL, awarded in 2018 for Marcon headquarters, in 2019 it was extended to the sites of Latisana and Montemarciano of San Marco Group S.p.A and to the Forlì-based Novacolor S.r.l. and, in 2020, also to the secondary production units incorporated into the San Marco Group. It obtained the Workplace safety recognition of safety management system as per the UNI INAIL (National Istitute for Insurance against Accidents at Work) model, for the Marcon headquarters.

San Marco Group received the Eurofins Indoor Air Comfort Gold certification, which is achieved if the products meet precise requirements for low VOC (Volatile Organic Compounds) emissions. In addition, the Marcon and Forlì sites have obtained REDcert^2 certification for a wide range of products, which is a certification schemes for the use of sustainable biomass in the chemical industry.

1.2 Environmental policy

San Marco Group offered water-based solutions for enamels and stains back in 1982 with the Unimarc Line and since then the pursuit has continued towards eliminating from its formulations raw materials considered hazardous to humans and the environment.

In 2010 San Marco Group has begun using the LCA methodology to understand the environmental performance of its products and to analyze their strengths and weaknesses. The holistic view of LCA convinced San Marco Group to acquire internally skills on the methodology and its application, with the ambitious goal of carrying out the LCA analysis of all the main products.

The LCA studies conducted enable San Marco Group to get a picture of its products from an environmental point of view, and to take actions of eco-design, both through actions for the improvement of its processes, and through the involvement of the supply chain in a virtuous circle. Moreover, the application of LCA in 2011 has allowed San Marco to achieve the EPD, or Environmental Product Declaration, for three products in the International EPD System.

In 2015 the thermal insulation system Marcotherm achieved the EPD.

In 2018 San Marco obtained the EPD process within the International EPD System which enables San Marco to create EPD autonomously. The EPD process is verified by a third-party verifier and, internally, once a year.

1.3 Product description

The object of this EPD is a is a lime-based finishing plaster coat, specially formulated for the decoration of prestigious interior and exterior, made of hydrated putty, fine marble powder, selected inert and rheological modifiers, which guarantees a perfect workability of the product. With its formulation, Teodorico permits to achieve a uniform and semi-polished finishing coat, similar to marble stone, creating soft and vibrant shades.

The analysis is in line with the PCR 2019:14 v1.3.4 "Construction products" [4] and the EN 15804:2012+A2:2019/AC2021 E [5].

The Teodorico is produced by San Marco (VE), which is part of San Marco Group, in the production plant located in the municipality of Marcon in the province of Venice and it is distributed by Novacolor, which is part of San Marco Group, located in the province of Forlì-Cesena.

The UN CPC code is 3511 "Paint and varnished and related products".

The product is available white in the package of 20.00 kg.

1.4 Content information

The Teodorico composition is shown in the following table:

		20 kg package		Declared Unit	Biogenic Material	
		Weight (kg)	%	Weight (kg)	Weight (kg C/kg e %)	
Bulk Product	Additives	0.516	2.5%	0.026	0.000	0.0%
	Fillers	13.162	63.6%	0.658	0.000	0.0%
	Dispersions and resins	1.071	5.2%	0.054	0.000	0.0%
	Salts and simple sustances	0.043	0.2%	0.002	0.000	0.0%
	Water	5.208	25.3%	0.260	0.000	0.0%
	Total bulk product	**20.000**	**96.7%**	**1.000**	**0.000**	**0.0%**
1° packaging	Polypropylene	0.480	2.3%	0.024	0.000	0.0%
	Recycled polypropylene	0.148	0.7%	0.007	0.000	0.0%
	Polyethylene	0.002	0.0%	0.000	0.000	0.0%
	Steel	0.057	0.3%	0.003	0.000	0.0%
	Total primary packaging	**0.687**	**3.3%**	**0.034**	**0.000**	**0.0%**
	Total	**20.687**	**100.0%**	**1.034**	**0.000**	**0.0%**

Table 1 – Teodorico

Teodorico product contains no SVHC (Substances of Very High Concern) on the Candidate List published by ECHA (European Chemicals Agency) in a concentration more than 0.1 % (w/w).

2. Environmental performance declaration

2.1 Declared unit

The declared unit is 1 kg of product with packaging.

The average consumption of the product Teodorico is 2.00-3.00 kg/m^2.

2.2 System boundaries

The present EPD is a declaration "from cradle to gate" (type "e" EPD). The system boundaries include raw material extraction, transport and production phase (A1-A2-A3).

The **Production phase** for San Marco manufacturing include:

- A1 - Raw materials and their packaging: raw materials extraction, production of product components and packaging, and energy;
- A2 – Transport: transportation of raw materials and packaging to San Marco plant;
- A3 – Manufacturing: washing operations, production and disposal of scrap, water consumption and waste treatment.

	Product stage			Construction process stage		Use stage							End of life stage					Resource recovery stage
	Raw materials	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal		Reuse-Recovery-Recycling-potential
Module	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4		D
Modules declared	x	x	x	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND		ND
Geography	GLO	GLO	IT	-	-	-	-	-	-	-	-	-	-	-	-	-		-
Specific data used	<10%																	
Variation – products	0%																	
Variation - sites	0%																	

Table 2 - Modules evaluated of the life cycle

No biogenic carbon content in the product and packaging[1].

[1] In the formulation of product there are biogenic raw materials, so the biogenic carbon has been balanced, because the amount is negligible (<1%)

2.3 Time boundaries

Primary data come from San Marco Group and refer to the year 2023. Secondary data come from the database ecoinvent v3.10 [6] released in 2023 and available in the LCA software used for calculations SimaPro 9.6 [7].

2.4 Geographic boundaries

The production site San Marco is located in the municipality of Marcon, in the province of Venice. The distribution site is Novacolor, which is part of San Marco Group and it is located in the province of Forlì-Cesena.

Raw materials come from Italy or from abroad.

2.5 Life cycle boundaries

In accordance with the options of the PCR for "Construction products", the following processes are excluded from the LCA: the construction of company buildings, the production of manufacturing equipment and other capital goods, personnel activities.

The contribution of infrastructure has not been excluded from the processes that already contained it, like the ecoinvent secondary processes.

For all raw materials modelled with stochiometric calc and literature data a conversion of 95% has been considered. For endothermic reaction or reaction in presence of heat, standard energy consumption of ecoinvent has been considered.

2.6 Boundaries to nature and other systems

Biogenic carbon removals and emissions are assessed with the Total Global Warming Potential (GWP-total) (including fossil GWP, biogenic GWP and land use GWP).

As required by the PCR 2019:14 v1.3.4, an additional mandatory indicator is included (GWP-GHG), adopting the carbon neutrality approach. This permits comparability with EPDs based on other PCRs, aligned with version 4.0 of the GPI [8].

2.7 Allocation and cut-off rules

As regards end-of-life allocation, the "polluter-pays" principle was adopted, particularly the "cut-off" approach. All unit processes before the point of end-of-waste shall be assigned to the product system generating the waste and all unit processes after the point of end-of-waste shall be assigned to the subsequent product system. Raw materials and manufacturing processes are included for virgin resources. For input recycled raw materials regranulation and remelting processes are considered, for output subject to recycling collecting, cleaning and sorting processes are included.

To calculate the production consumption of 1 kg of product, the annual factory consumption and waste of San Marco in Marcon (VE) plant in 2023 were attributed on a mass basis, considering the total quantity of all products manufactured.

No cut-off rules were applied.

No co-products allocation was applied because there are not co-products.

2.8 Database and LCA software used

Ecoinvent 3.10., SimaPro 9.6

3. Data quality

Primary data have been used for the fundamental aspects of the study, such as energy and water consumption in production plants of San Marco or the product composition. For all processes for which primary or representative data were not available, the LCA database ecoinvent v3.10, cut-off by classification, was used.

Data quality has been evaluated through the pedigree matrix for all processes with incidence >2% for the main LCIA categories and a qualitative analysis for remaining processes. A sensitivity analysis of proxy data has been carried out to verify the contribution to the main LCIA categories is lower than 10% as required by the reference PCR.

All inputs of the productive process have been evaluated.

For data collection specific questionnaire has been used, San Marco Group has been provided safety and technical data sheet for raw materials and products, certification and a company consumption excel file. All data from questionnaire has been included in a file, the file has been imported in the tool in SimaPro.

Raw materials not available in the ecoinvent database have been modelled from the information contained in the respective safety and technical data sheets or by modifying existing ecoinvent processes of similar chemicals or with stoichiometric calculation.

For San Marco plant electricity consumption, a residual electricity mix (Electricity, medium voltage {IT}| electricity, medium voltage, residual mix | Cut-off, S) as purchased and Electricity, low voltage {IT}| electricity production, photovoltaic, 3kWp slanted-roof installation, multi-Si, panel, mounted | Cut-off, S (self-production). The GWP-GHG indicator related to the electricity mix is equal to 0.635 kg CO2 eq/kWh.

To define the plant consumption (A3) data concerning the consumption of electricity, water and waste production in 2023 have been used.

4. Environmental impact

4.1 Environmental impact indicators

The following environmental impact indicators are considered, according to PCR 2019:14 v1.3.4 and UNI EN EN 15804:2012+A2:2019/AC2021 E.

Nr.	Core environmental impact indicators	Unit
1.1	Global Warming Potential - total (GWP-total)	kg CO2 eq.
1.2	Global Warming Potential - fossil fuels (GWP-fossil)	kg CO2 eq.
1.3	Global Warming Potential - biogenic (GWP-biogenic)	kg CO2 eq.
1.4	Global Warming Potential - land use and land use change (GWP-luluc)	kg CO2 eq.
1.5	Depletion potential of the stratospheric ozone layer (ODP)	kg CFC-11 eq.
1.6	Acidification potential, Accumulated Exceedance (AP)	mol H+ eq.
1.8	Eutrophication potential - freshwater (EP-freshwater)	kg P eq
1.9	Eutrophication potential - marine (EP-marine)	kg N eq.
1.10	Eutrophication potential - terrestrial (EP-terrestrial)	mol N eq.
1.11	Photochemical Ozone Creation Potential (POCP)	kg NMVOC eq.
1.12	Abiotic depletion potential - non-fossil resources (ADPE)	kg Sb eq.
1.13	Abiotic depletion potential - fossil resources (ADPF)	MJ
1.14	Water (user) deprivation potential (WDP)	m3 world eq. deprived
	Additional mandatory environmental impact indicators	
2.1.	Global Warming Potential (GWP-GHG)	kg CO2 eq.
	Additional voluntary environmental impact indicators	
3.1	Particulate Matter emissions (PM)	Disease incidence
3.2	Ionizing radiation, human health (IRP)	kBq U235 eq.
3.3	Eco-toxicity - freshwater (ETP-fw)	CTUe
3.4	Human toxicity, cancer effect (HTP-c)	CTUh
3.5	Human toxicity, non-cancer effects (HTP-nc)	CTUh
3.6	Land use related impacts/Soil quality (SQP)	dimensionless
	Indicators describing resource use	
4.1	Use of renewable primary energy as energy carrier (PERE)	MJ
4.2	Use of renewable primary energy resources used as raw materials (PERM)	MJ
4.3	Total use of renewable primary energy (PERT)	MJ
4.4	Use of non-renewable primary energy as energy carrier (PENRE)	MJ

4.5	Use of non-renewable primary energy resources used as raw materials (PENRM)	MJ
4.6	Total use of non-renewable primary energy resource (PENRT)	MJ
4.7	Use of secondary material (SM)	kg
4.8	Use of renewable secondary fuels (RSF)	MJ
4.9	Use of non-renewable secondary fuels (NRSF)	MJ
4.10	Net use of fresh water (FW)	m3
	Environmental information describing waste categories	
5.1	Hazardous waste disposed (HWD)	kg
5.2	Non-hazardous waste disposed (NHWD)	kg
5.3	Radioactive waste disposed (RWD)	kg
	Environmental information describing output flows	
6.1	Components for re-use (CRU)	kg
6.2	Materials for recycling (MFR)	kg
6.3	Materials for energy recovery (MER)	kg
6.4	Exported energy (EE), electricity	MJ
6.5	Exported energy (EE), thermal	MJ

Table 3 - Environmental indicators included in the analysis.

4.2 Environmental impact assessment of Teodorico

Potential environmental impact – mandatory indicators according to EN 15804

	Results for declared unit	
Indicator	**Unit**	**A1-A2-A3**
GWP-total	kg CO2 eq.	5.73E-01
GWP-fossil	kg CO2 eq.	5.70E-01
GWP-biogenic	kg CO2 eq.	2.93E-03
GWP-luluc	kg CO2 eq.	2.80E-04
ODP	kg CFC 11 eq.	1.11E-08
AP	mol H+eq.	1.57E-03
EP-freshwater	kg P eq	8.73E-05
EP-marine	kg N eq.	3.71E-04
EP-terrestrial	mol N eq.	3.84E-03
POCP	kg NMVOC eq.	1.92E-03
ADP-minerals & metals*[2]	kg Sb eq.	2.13E-06
ADP-fossil*	MJ	8.92E+00
WDP*	m3 world eq. deprived	1.12E-01
Acronyms		GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption

Table 4 - Results for declared unit

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.*

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Potential environmental impact – additional mandatory and voluntary indicators

	Results for declared unit	
Indicator	**Unit**	**A1-A2-A3**
GWP-GHG[2]	kg CO2 eq.	5.71E-01
PM	Disease incidence	1.87E-08
IRP**	kBq U235 eq.	3.46E-02
ETP-fw*[2]	CTUe	2.40E+00
HTP-c*[2]	CTUh	2.32E-09
HTP-nc*[2]	CTUh	5.09E-09
SQP*[3]	dimensionless	3.00E+00
Acronyms	GWP-GHG = Global Warming Potential neutral; PM = Potential incidence of disease due to PM emissions; IRP = Potential human exposure efficiency relative to U235, ETP-fw = Potential comparative toxic unit for ecosystems; HTP-c = Potential comparative toxic unit for effects; HTP-nc = Potential comparative toxic unit for humans; SQP = Potential soil quality index.	

Table 5 – Results for declared unit - additional indicator

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

*** Disclaimer: This impact category deals mainly with the eventual impact of low dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities, Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.*

[2]The indicator includes all greenhouse gases included in GWP-total but excludes biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product

[3] The results of the impact categories abiotic depletion of minerals and metals, land use, human toxicity (cancer), human toxicity (non-cancer) and ecotoxicity (freshwater) may be highly uncertain in LCAs that include capital goods/infrastructure in generic datasets, in case infrastructure/capital goods contribute greatly to the total results. This is because the LCI data of infrastructure/capital goods used to quantify these indicators in currently available generic datasets sometimes lack temporal, technological and geographical representativeness. Caution should be exercised when using the results of these indicators for decision-making purposes.

Use of resources

	Results for declared unit	
Indicator	**Unit**	**A1-A2-A3**
PERE	MJ	6.54E-01
PERM	MJ	0.00E+00
PERT	MJ	6.54E-01
PENRE	MJ	7.88E+00
PENRM	MJ.	1.04E+00
PENRT	MJ	8.92E+00
SM	kg	7.45E-03
RSF	MJ	0.00E+00
NRSF	MJ	0.00E+00
FW	m3	3.27E-03
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water	

Table 6 - Use of resources

Waste production and output flows

Waste production

	Results for declared unit	
Indicator	**Unit**	**A1-A2-A3**
Hazardous waste disposed	kg	9.67E-05
Non-hazardous waste disposed	kg	8.24E-02
Radioactive waste disposed	kg	8.75E-06

Table 7 - Waste production

Output flows

Results for declared unit		
Indicator	**Unit**	**A1-A2-A3**
Components for re-use	kg	0.00E+00
Material for recycling	kg	0.00E+00
Materials for Energy recovery	kg	0.00E+00
Exported energy, electricity	MJ	7.91E-03
Exported energy, thermal	MJ	1.83E-02

Table 8 - Output flows

5. Additional environmental information

5.1 Certification

The product is certified “Indoor Air Comfort Gold”. This certificate ensures that low product emission requirements are fulfilled and it is a sign of the applicant’s focus on quality and contribution to a healthy indoor environment.

Certificate number: IACG-442-01-02-2021, Validity date: 2026-12-03

6. General information

6.1 San Marco Group Spa information

The Life Cycle Assessment (LCA) study and this EPD were created by the Product Safety Department of San Marco Group SpA, in collaboration with 2B Srl (www.to-be.it).

The company contact details are:

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6.2 Program information

Programme EPD:	The International EPD® System
Address	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website	www.environdec.com
E-mail	info@environdec.com

Accountabilities for PCR, LCA and independent, third-party verification
Product Category Rules (PCR)
CEN standard EN 15804 serve as the core Product Category Rules (PCR)
Product category rules (PCR): PCR 2019:14 Construction products, version 1.3.4
PCR review was conducted by: The Technical Committee of the International EPD® System. See www.environdec.com/TC for a list of members. Review chair: Claudia A. Peña, University of Concepción, Chile. The review panel may be contacted via the Secretariat www.environdec.com/contact.
Life cycle assessment (LCA)
LCA accountability: 2B Srl
Third-party verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006 via: ☑ EPD verification by EPD Process Certification* Internal auditor: Leo Breedveld, 2B Srl Third-party verification: CSQA Certificazioni Srl is an approved certification body accountable for third-party verification Third-party verifier is accredited by: Accredia (00070) *For EPD Process Certification, an accredited certification body certifies and reviews the management process and verifies EPDs published on a regular basis. For details about third-party verification procedure of the EPDs, see the GPI.
Procedure for follow-up of data during EPD validity involves third-party verifier? ☑ Yes ☐ No

San Marco Group SpA is the exclusive owner of the EPD and is responsible for its content.

EPDs within the same product category but registered in different EPD programmes may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison.

7. Bibliography

[1] San Marco Group Spa (www.sanmarcogroup.it).

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[7] PRé Sustainability, Software SimaPro 9.6, 2023 (www.simapro.com).

[8] EPD International, General Programme Instructions for the International EPD System. Version 4.0, 2021.